

# LegalEase AI - Development Setup Guide

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## Overview

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This guide provides the exact step-by-step instructions to set up, configure, and run **LegalEase AI** on Windows and within the Antigravity workspace.

## Prerequisites

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- **Python:** Version 3.10+ (Tested on Python 3.14.6)
- **Pip:** Latest version
- **Google Gemini API Key:** Free tier or paid key from [Google AI Studio](#) 

## Step 1 — Verify Python Installation

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Open PowerShell and verify Python is installed:

```
powershell  
python --version
```

Expected output:

```
text  
Python 3.10+ (or Python 3.14.x)
```

## Step 2 — Create Virtual Environment

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Navigate to the project root directory:

```
powershell  
cd C:\Users\DELL\.gemini\antigravity\scratch\LegalEase-AI  
python -m venv venv
```

## Step 3 — Activate the Virtual Environment

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```
powershell
venv\Scripts\activate
```

Upon activation, your shell prompt will show `(venv)` .

## Step 4 — Install Required Dependencies

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Upgrade `pip` and install all project packages:

```
powershell
python -m pip install --upgrade pip
pip install -r requirements.txt
```

Verified packages installed:

- `fastapi` & `uvicorn` (REST API & ASGI server)
- `streamlit` (Web UI frontend)
- `requests` (Frontend HTTP client)
- `pydantic` (Data validation)
- `python-dotenv` (Environment configuration)
- `google-genai` & `google-generativeai` (Gemini SDKs)
- `python-docx` (Microsoft Word generator)
- `reportlab` (PDF document generator)
- `Pillow` (Image verification & logo generation)
- `pytest` & `httpx` (Automated testing suite)

## Step 5 — Configure Environment Variables (.env)

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Copy `.env.example` to `.env` :

```
powershell
copy .env.example .env
```

Open `.env` in an editor and enter your Gemini API key:

```
env
# Google Gemini API Key
GEMINI_API_KEY=your_gemini_api_key_here

# Gemini Model Selection
GEMINI_MODEL=gemini-2.5-flash

# Backend Network Settings
BACKEND_HOST=0.0.0.0
BACKEND_PORT=8000
BACKEND_URL=http://localhost:8000
```

## Step 6 — Start the FastAPI Backend

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Launch the backend server in a terminal window:

```
powershell
uvicorn backend.main:app --reload --port 8000
```

Expected startup log:

```
text
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:      Application startup complete.
```

- Verify API root: `http://localhost:8000/`
- Verify Swagger UI: `http://localhost:8000/docs`

## Step 7 — Start the Streamlit Frontend

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Open a second PowerShell window, activate `venv`, and start Streamlit:

```
powershell
venv\Scripts\activate
streamlit run frontend/app.py
```

Expected startup log:

text

You can now view your Streamlit app in your browser.

Local URL: `http://localhost:8501`

Network URL: `http://192.168.x.x:8501`

Open your browser to `http://localhost:8501`.

## Quick One-Click Windows Launcher

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Alternatively, run the provided batch script to launch both services simultaneously:

```
powershell  
.\run_project.bat
```